Clustering

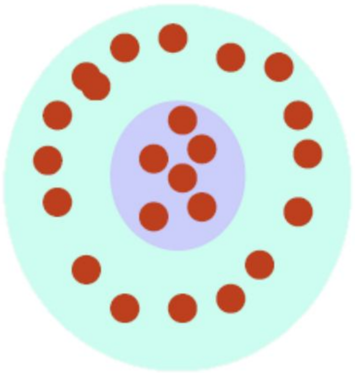
Arin Ghazarian
California State University Long Beach

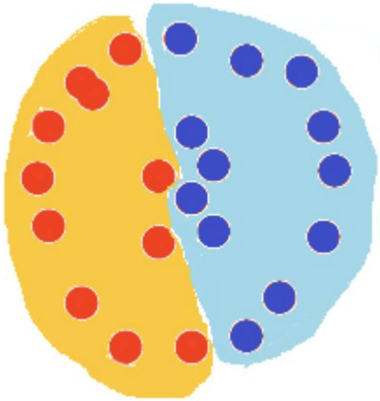
Unsupervised Learning

- Finding structure in unlabeled data
- Data has no label
- In fact, we want to assign labels (cluster IDs)

Clustering

- The organization of unlabeled data into similarity groups (clusters)
- Data points inside the same cluster are “similar”, and “dissimilar” to data points in other clusters.





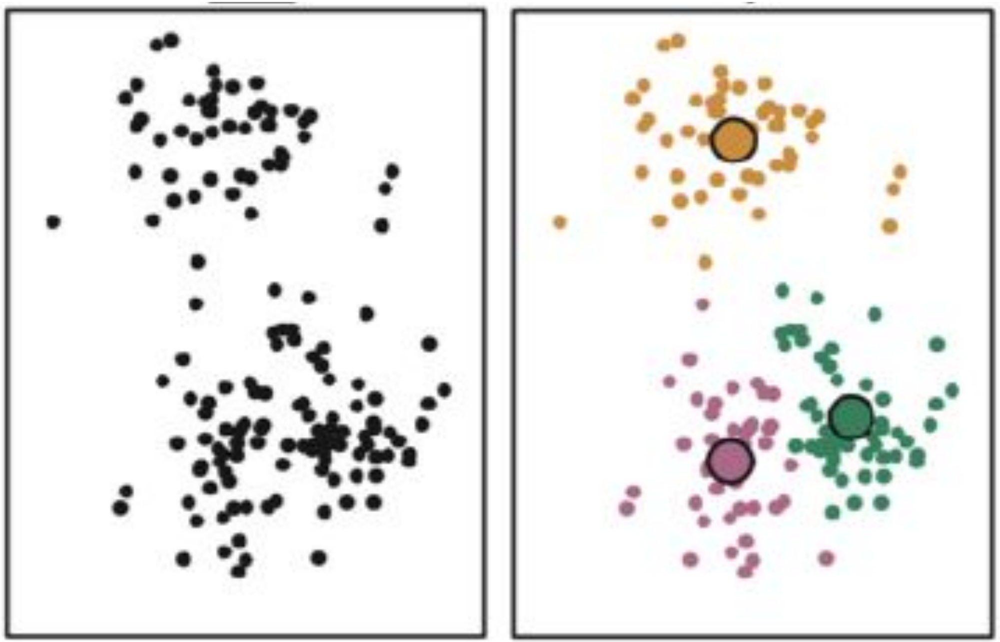
http://www.mit.edu/~9.54/fall14/slides/Class13.pdf

a b

https://link.springer.com/article/10.1007/s11042-021-10594-9

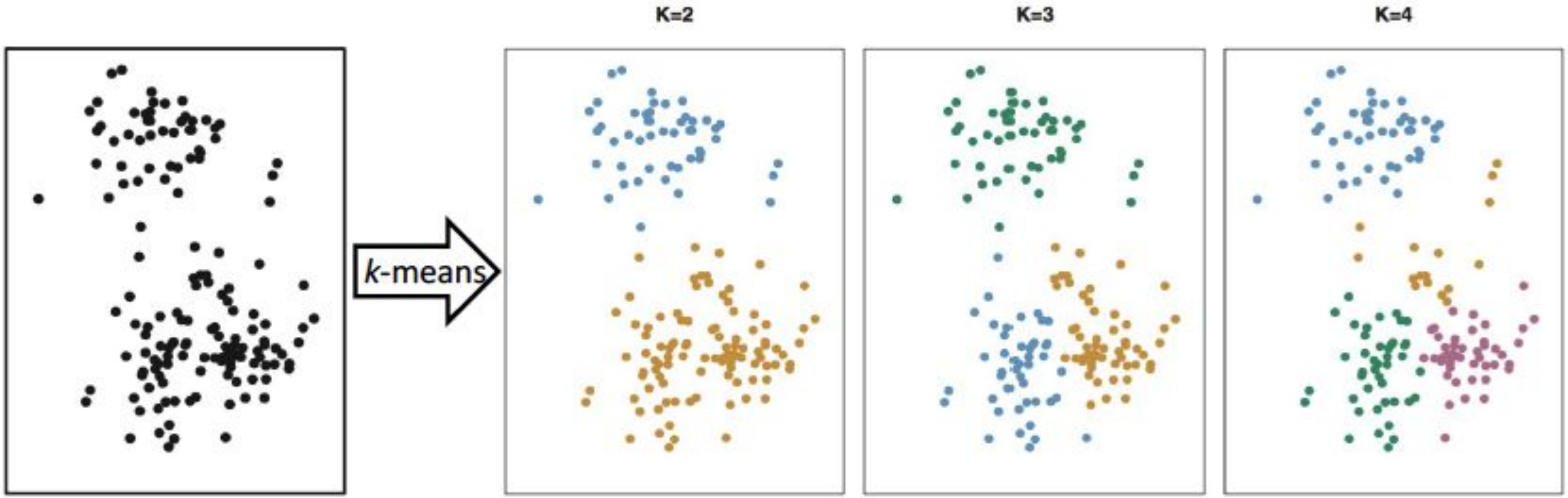
Clustering

- This can help explain data that might otherwise appear unrelated and to have a more-concise representation of the data.



K-Means Clustering

- Groups data based on some measure of proximity
- Which samples belong together?
- What value should we pick for K?



Formulating K-means

n dimensional space

$$X = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\},$$

k clusters

$$\{C_1, \dots, C_k\}$$

Partitioning n data points into k clusters

$$C_1 \cup C_2 \cup \cdots \cup C_k = X$$

$$C_i \cap C_j = \emptyset \ \ \forall i, j$$

Formulating K-means

We want to minimize the variation within each cluster (we use squared Euclidean distance)

$$W(C_p) = \frac{1}{|C_p|} \sum_{i,j \in C_p} (\vec{x}_i - \vec{x}_j)^2 .$$

For all clusters, we minimize the sum of within-cluster variation for all k clusters

$$\min_{C_1,\ldots,C_k} \left\{ \sum_{p=1}^{k} W(C_p) \right\} = \min_{C_1,\ldots,C_k} \left\{ \sum_{p=1}^{k} \frac{1}{|C_p|} \sum_{i,j \in C_p} (\vec{x}_i - \vec{x}_j)^2 \right\} .$$

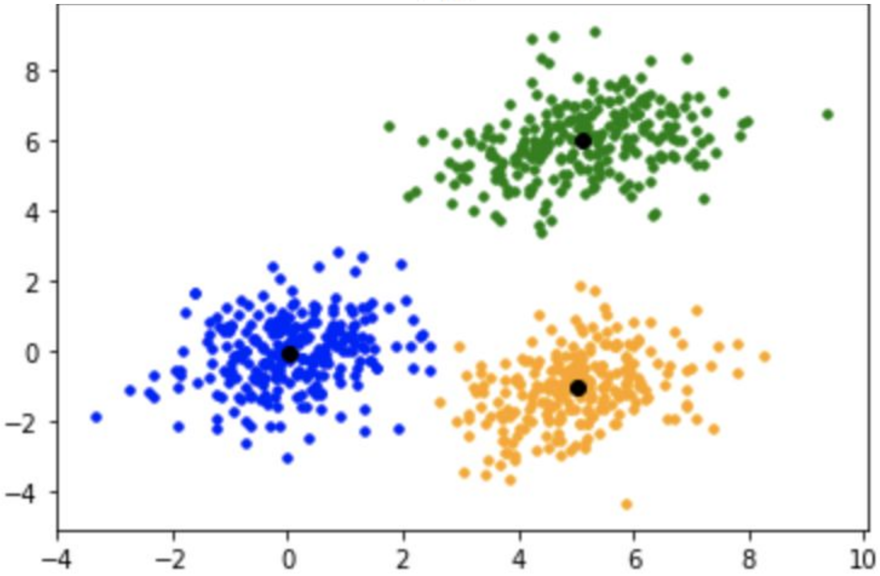
Formulating K-means

- We can not find the global minimum
- Instead we use a local minimum

$$\min_{C_1,\ldots,C_k}\left\{\sum_{p=1}^{k} W(C_p)\right\} = \min_{C_1,\ldots,C_k}\left\{\sum_{p=1}^{k}\frac{1}{|C_p|}\sum_{i,j\in C_p}(\vec{x}_i-\vec{x}_j)^2\right\}.$$

Cluster centroid

$$\overline{\vec{x}_{C_p}} = \frac{1}{|C_p|} \sum_{i \in C_p} \vec{x}_i \,.$$

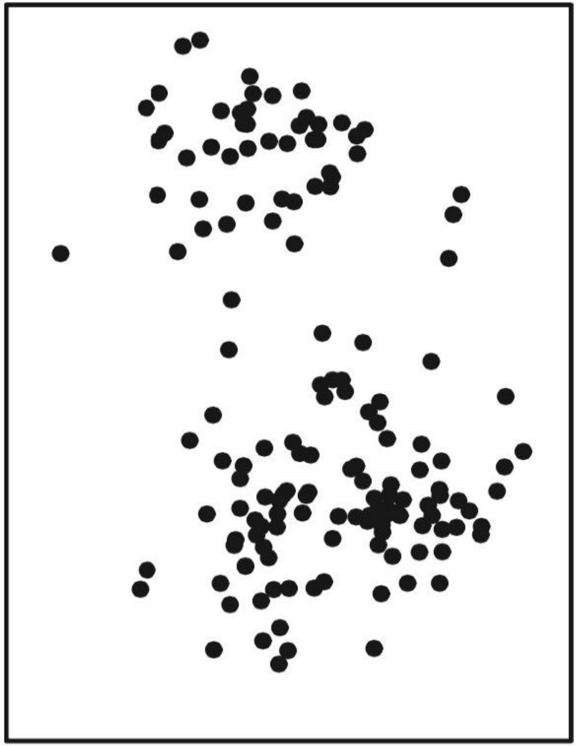


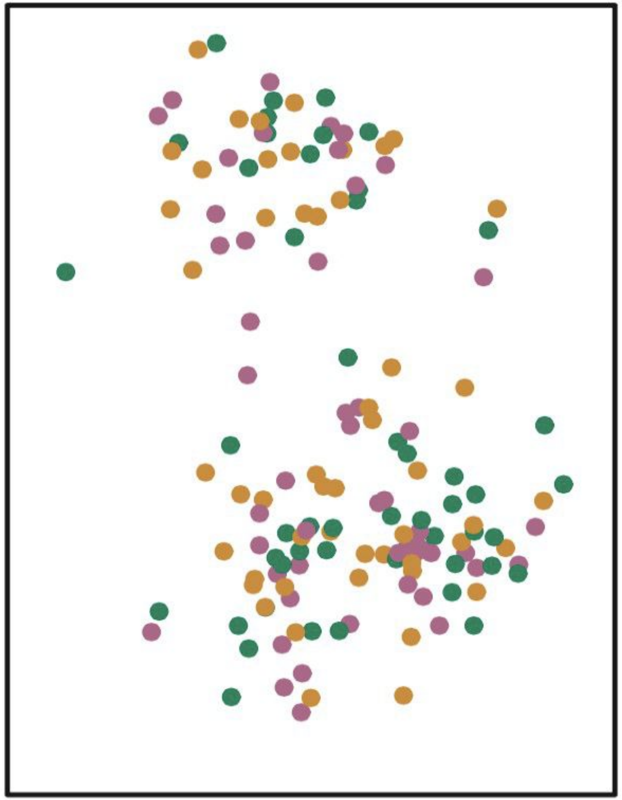
https://www.geeksforgeeks.org/ml-k-means-algorithm/

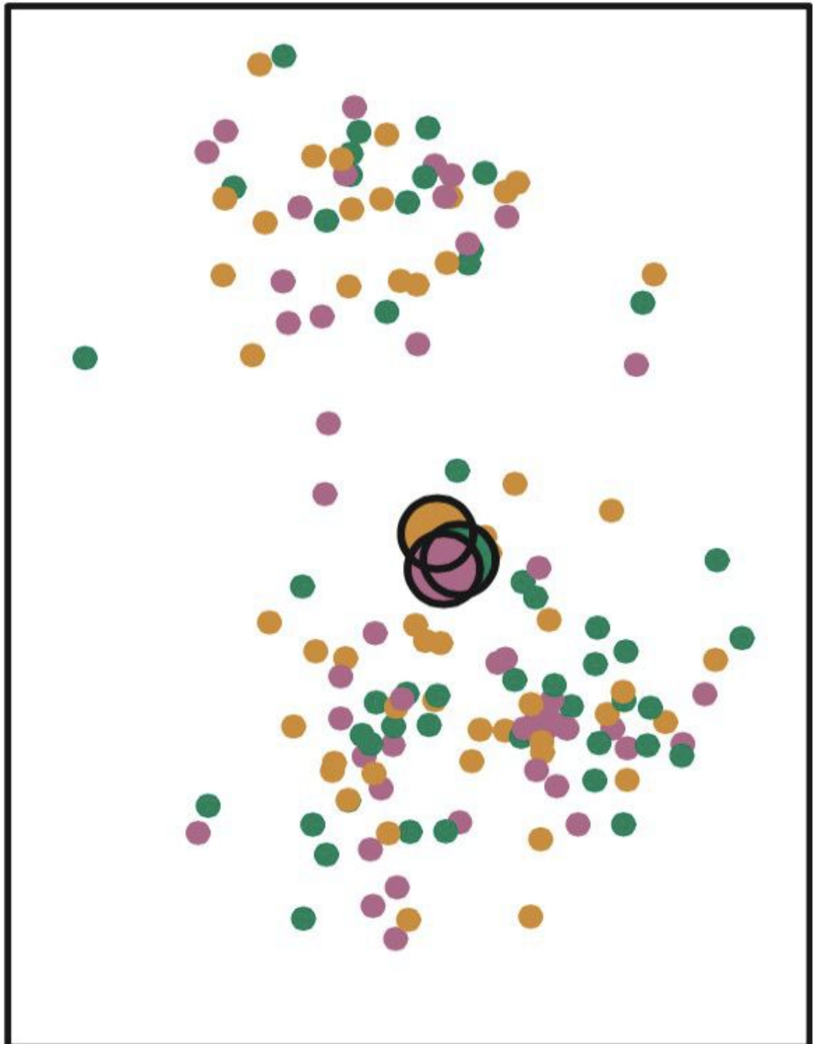
K-Means Algorithm

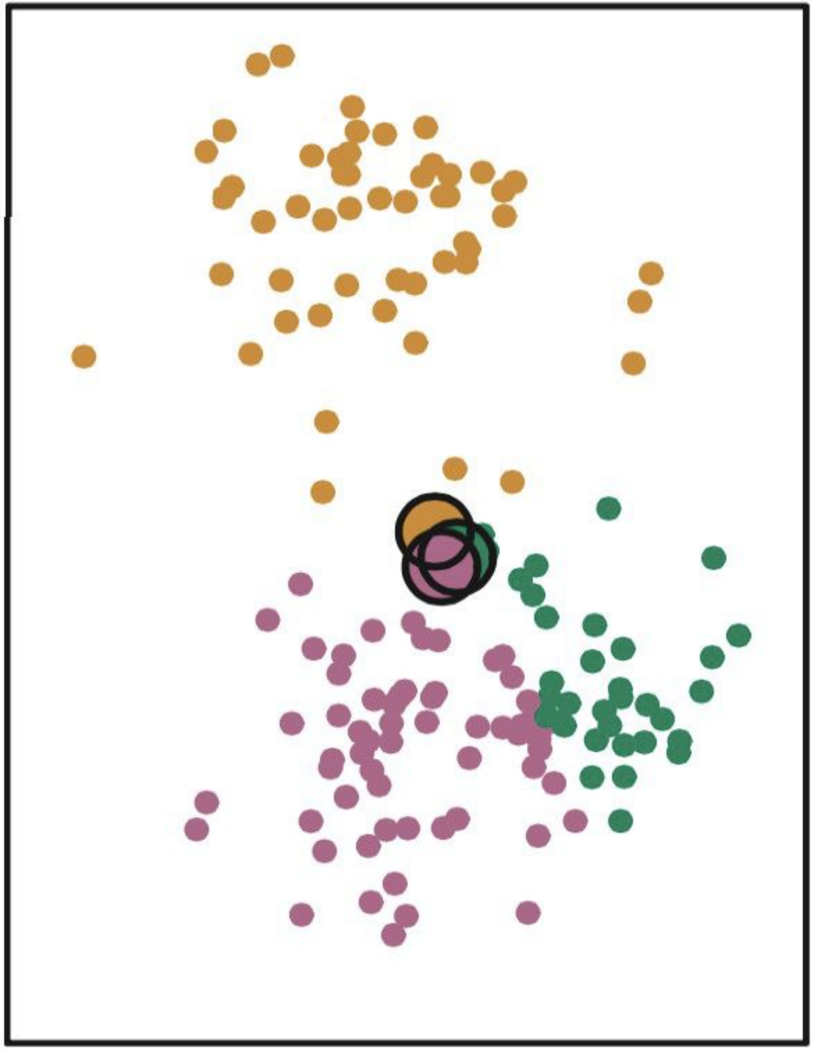
1. Pick k (the number of clusters)
2. Randomly assign each sample to a cluster
3. Repeat until the clusters stabilize:
    a. Compute each cluster’s centroid
    b. Assign each sample to the cluster to whose centroid it is closest (using Euclidean or other distance measures)

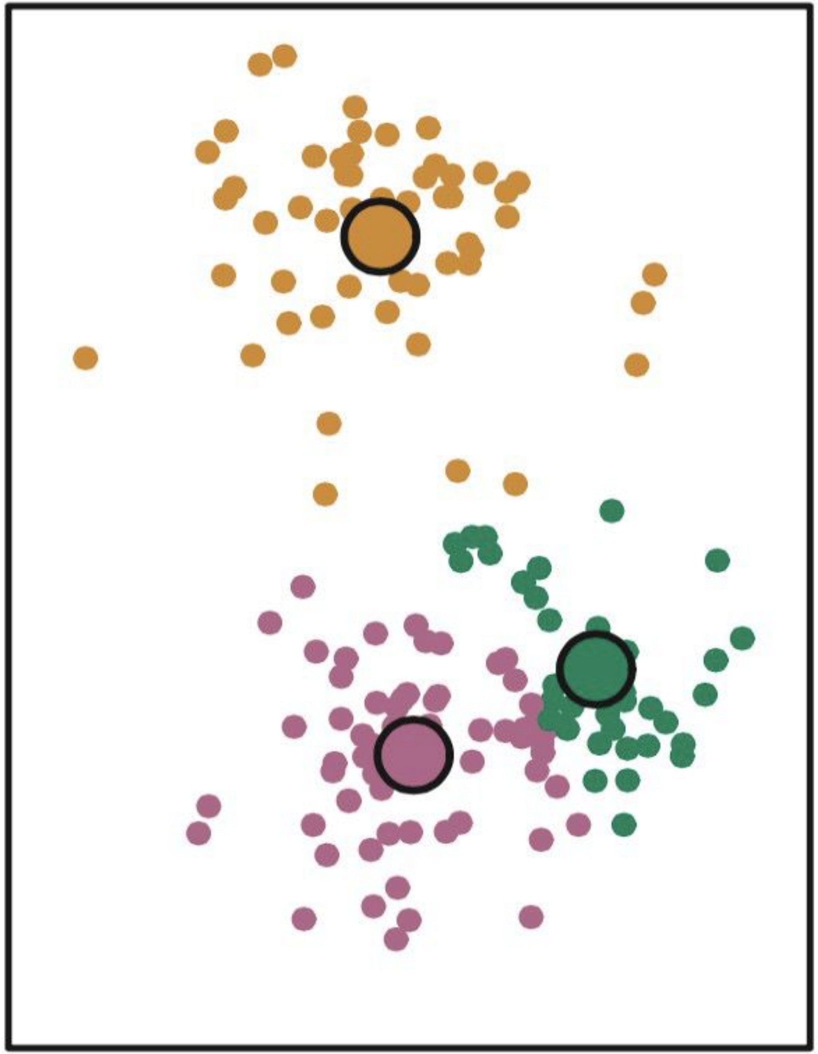
Example



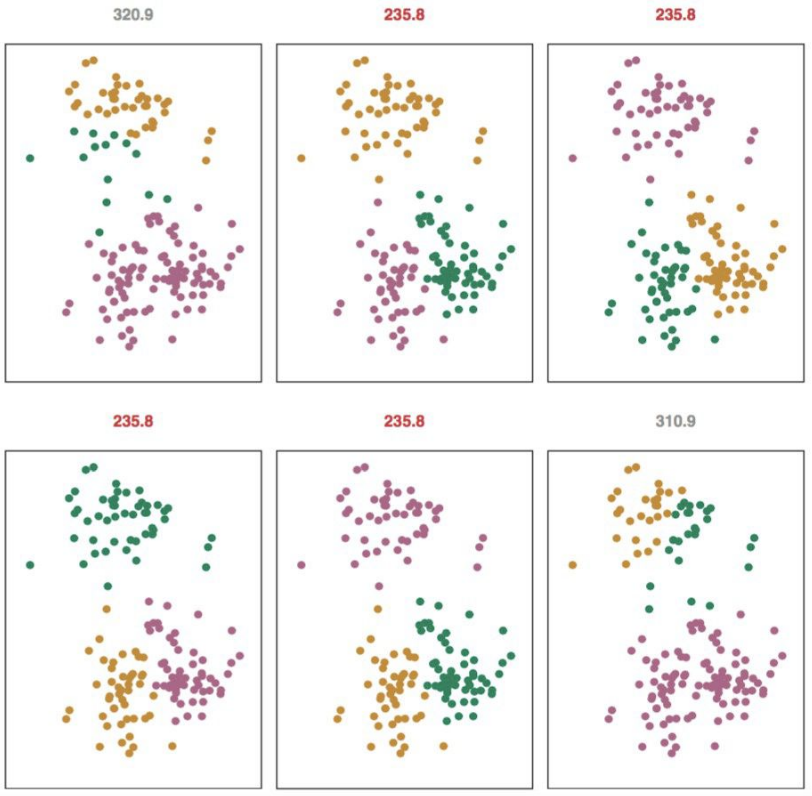






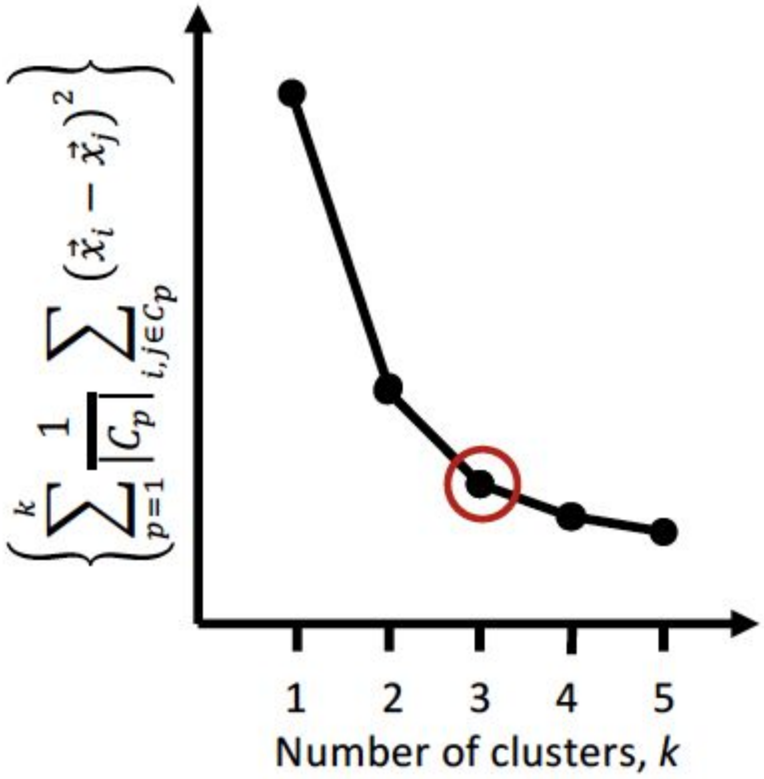


- We typically run k-means several times with a different random initial clustering
- This often leads to different results
- We pick the one with the minimal within-cluster variability



How to choose k?

- Elbow Method
  - The elbow is where the decrease in the plot begins to diminish.
  - Subjective



How to choose K?

- The silhouette coefficient
- For each sample:

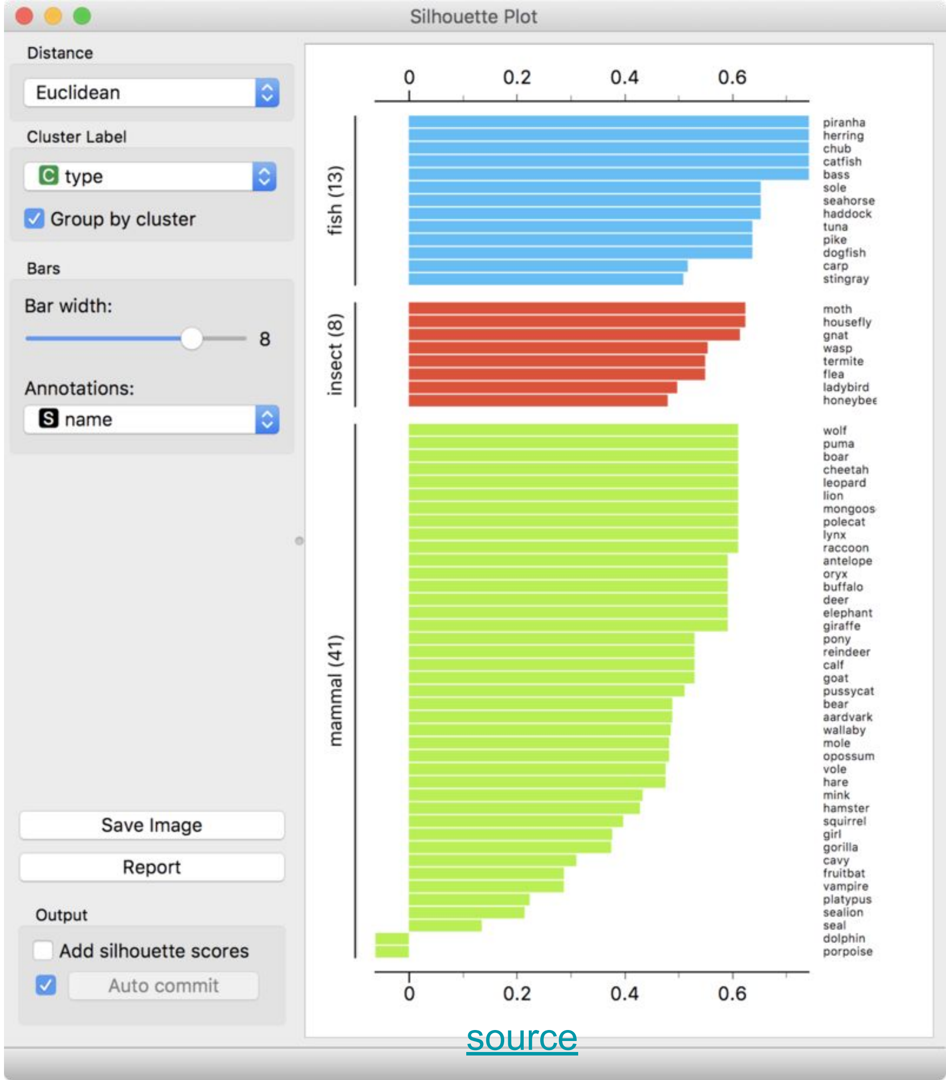
$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

- Here, a(i) is the average dissimilarity between sample i and all other samples in i’s own cluster.
- b(i) is the lowest average dissimilarity between i and any other cluster
- $-1 <= s(i) <= 1$ (we want it to be close to 1)
- The silhouette can be calculated with any distance metric, such as the Euclidean distance or the Manhattan distance.
- Silhouette coefficients values
    - Near +1 → the sample is far away from the neighboring clusters. A
    - Near 0 → the sample is very close to the decision boundary between two neighboring clusters
    - Negative values → samples might have been assigned to the wrong cluster

Silhouette coefficient method

- The mean s(i) over all points of a cluster is a measure of how tightly grouped all the points in the cluster are.
- Thus the mean s(i) over all data of the entire dataset is a measure of how appropriately the data have been clustered.
- If there are too many or too few clusters, as may occur when a poor choice of k is used in the clustering algorithm, some of the clusters will typically display much narrower silhouettes than the rest.

A plot showing silhouette scores from three types of animals from the Zoo data set as rendered by Orange data mining suite. At the bottom of the plot, silhouette identifies dolphin and porpoise as outliers in the group of mammals.

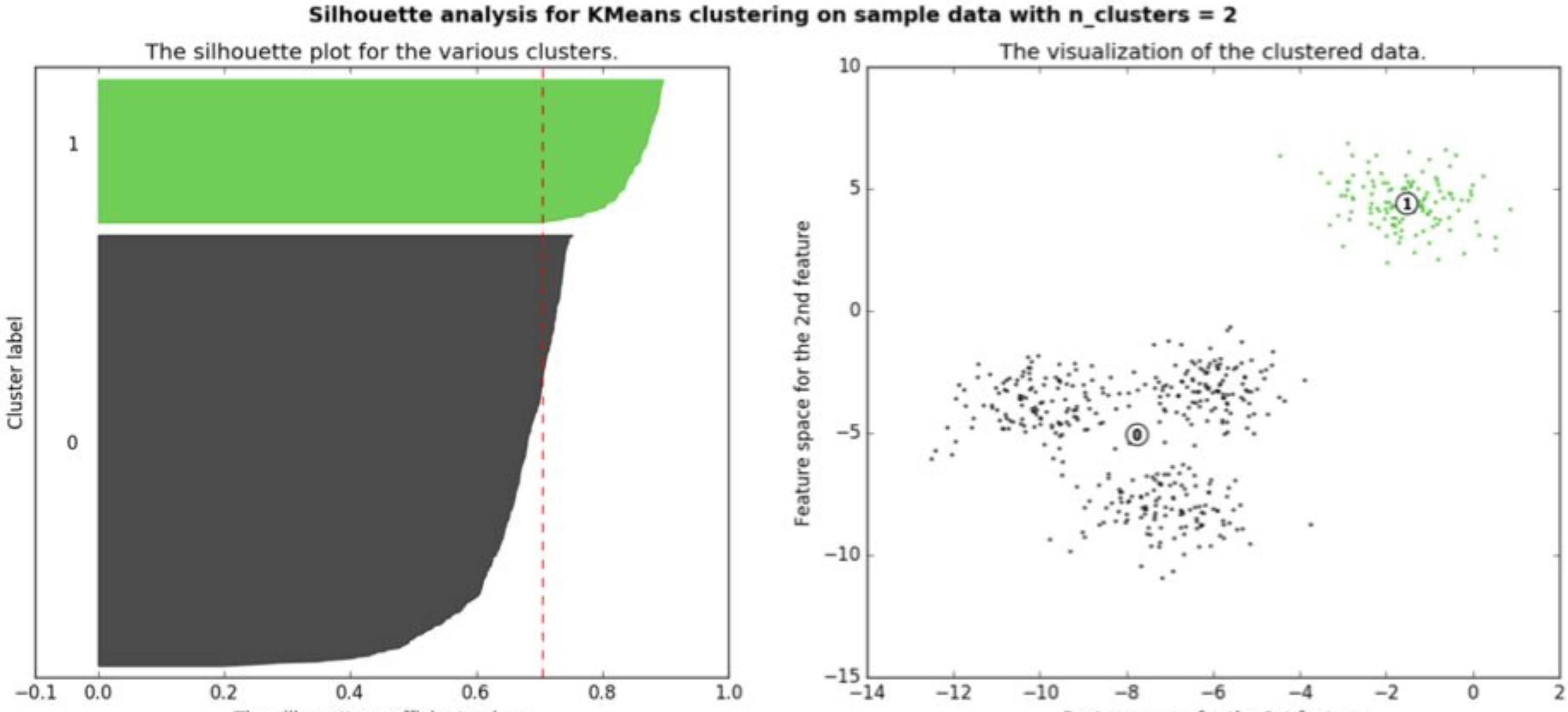


source

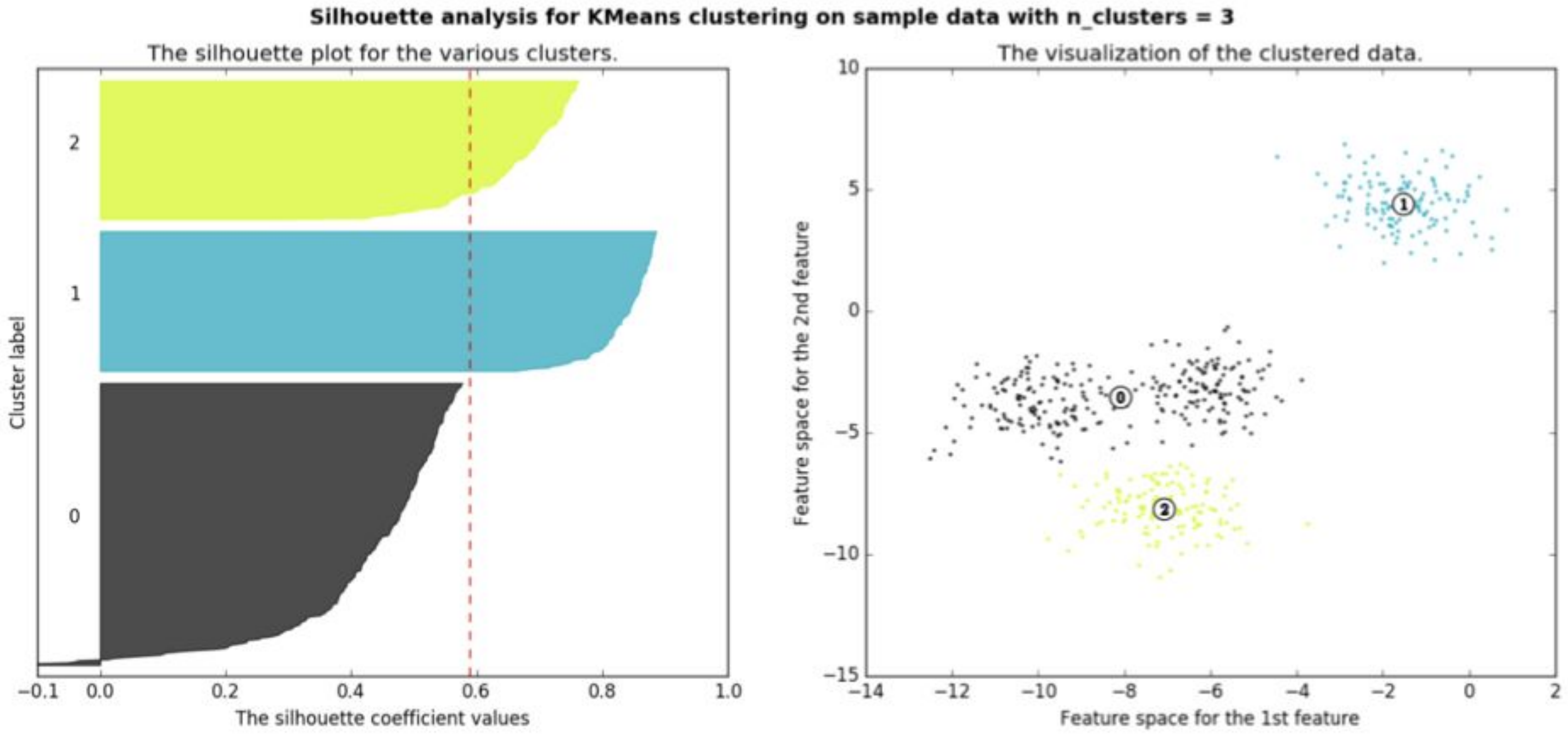
k=2

Bad option for k:

- Presence of clusters with below average silhouette scores
- Wide fluctuations in the size of the silhouette plots

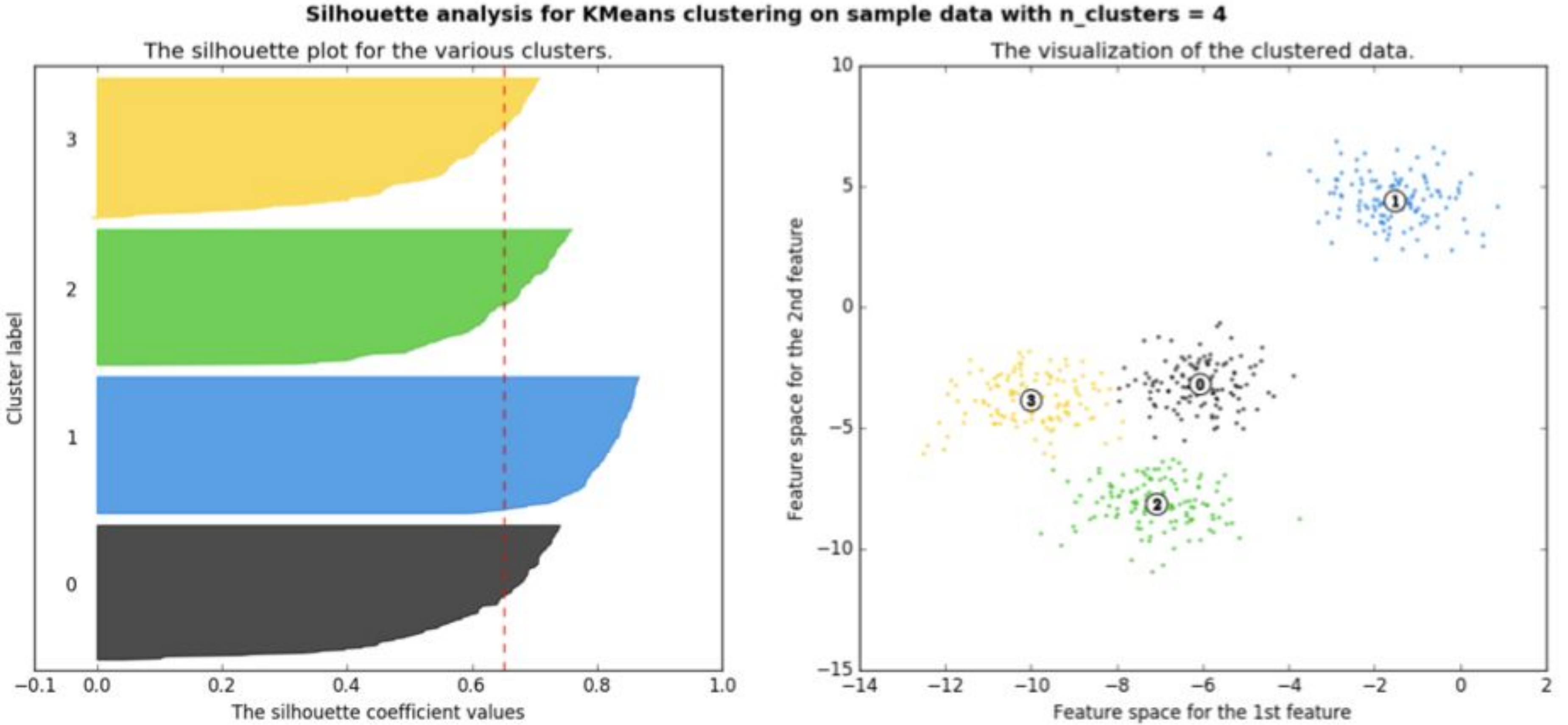


k=3

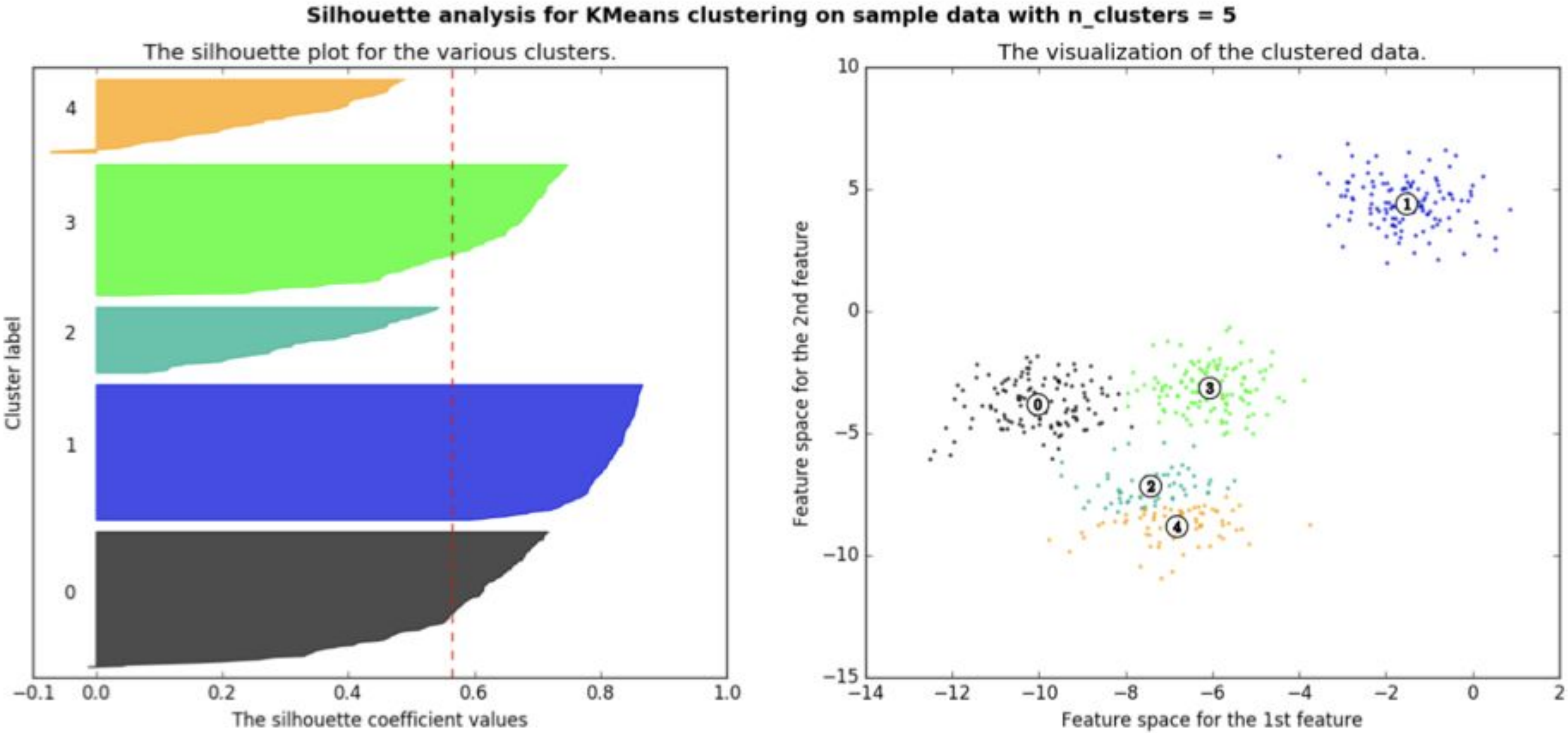

Silhouette analysis for KMeans clustering on sample data with n_clusters = 3
The silhouette plot for the various clusters.
Cluster label
2
1
0
−0.1
0.0
0.2
0.4
0.6
0.8
1.0
The silhouette coefficient values
The visualization of the clustered data.
Feature space for the 2nd feature
10
5
0
−5
−10
−15
−14
−12
−10
−8
−6
−4
−2
0
2
Feature space for the 1st feature

k=4

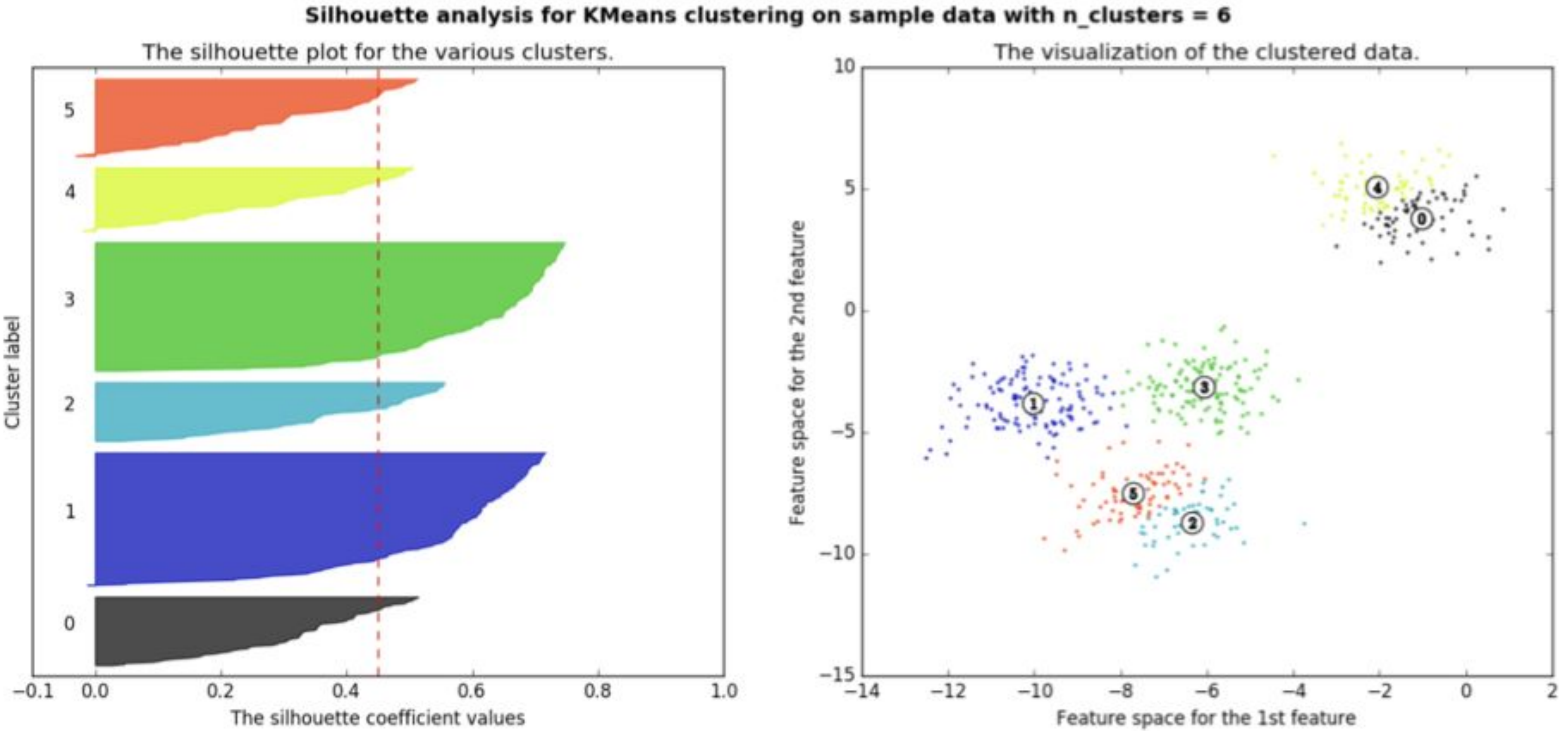
The thickness of clusters is more uniform than here



k=5

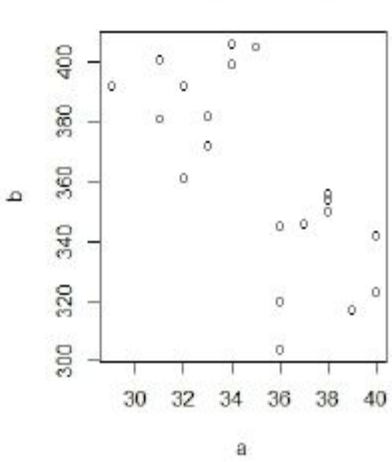

Silhouette analysis for KMeans clustering on sample data with n_clusters = 5
The silhouette plot for the various clusters.
The visualization of the clustered data.
Cluster label
The silhouette coefficient values
Feature space for the 1st feature
Feature space for the 2nd feature
−0.1
0.0
0.2
0.4
0.6
0.8
1.0
−14
−12
−10
−8
−6
−4
−2
0
2
10
5
0
−5
−10
−15
0
1
2
3
4

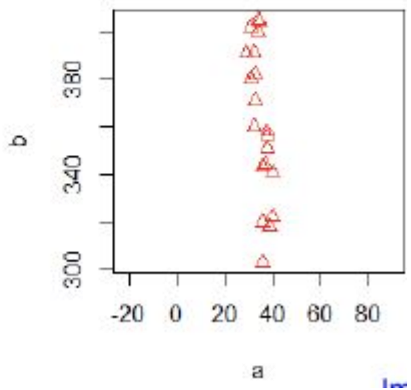
k=6


Silhouette analysis for KMeans clustering on sample data with n_clusters = 6
The silhouette plot for the various clusters.
Cluster label
The silhouette coefficient values
−0.1 0.0 0.2 0.4 0.6 0.8 1.0
0 1 2 3 4 5
The visualization of the clustered data.
Feature space for the 1st feature
Feature space for the 2nd feature
−14 −12 −10 −8 −6 −4 −2 0 2
10 5 0 −5 −10 −15

Clustering is not scale invariant

We might need to normalize data

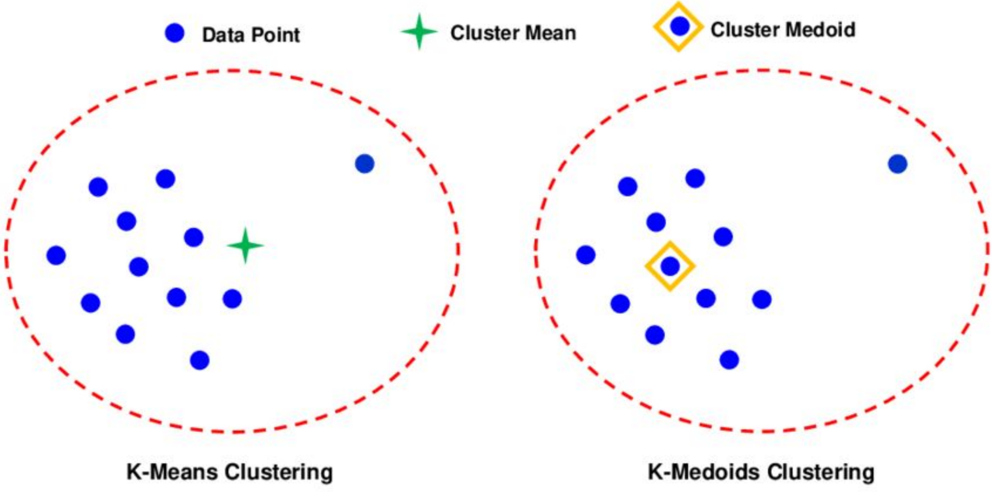




K-medoids

- The k-means clustering uses centroids
- Centroids are not robust to outliers

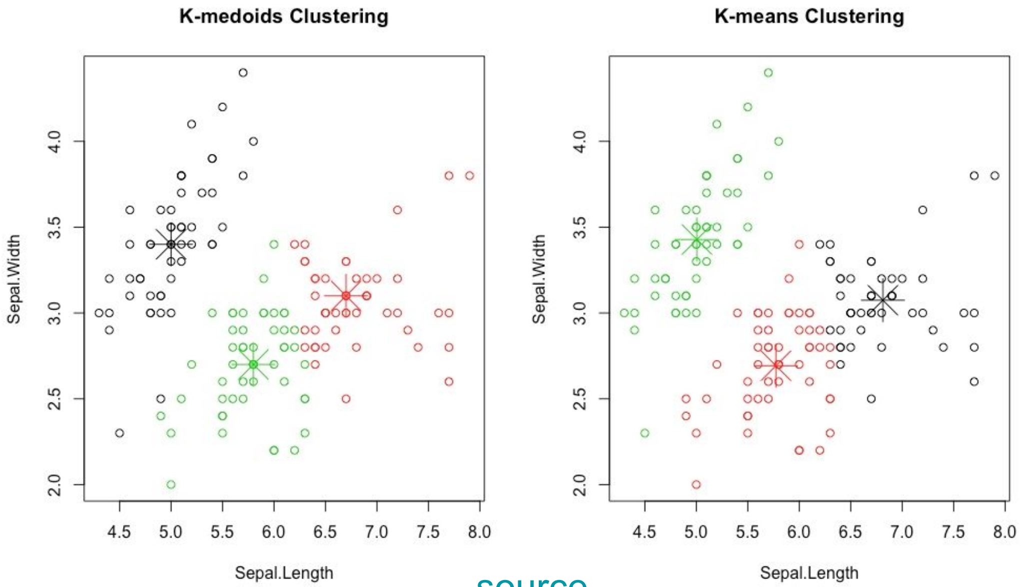
$$\overline{\vec{x}_{C_p}} = \frac{1}{|C_p|} \sum_{i \in C_p} \vec{x}_i .$$



https://www.researchgate.net/figure/The-graphical-representation-of-the-difference-between-the-k-means-and-k-medoids_fig1_342871651

K-medoids

- We can use median of a cluster instead which is robust to outliers
- But k-means is O(kn) while k-medoids might be O(kn^2).



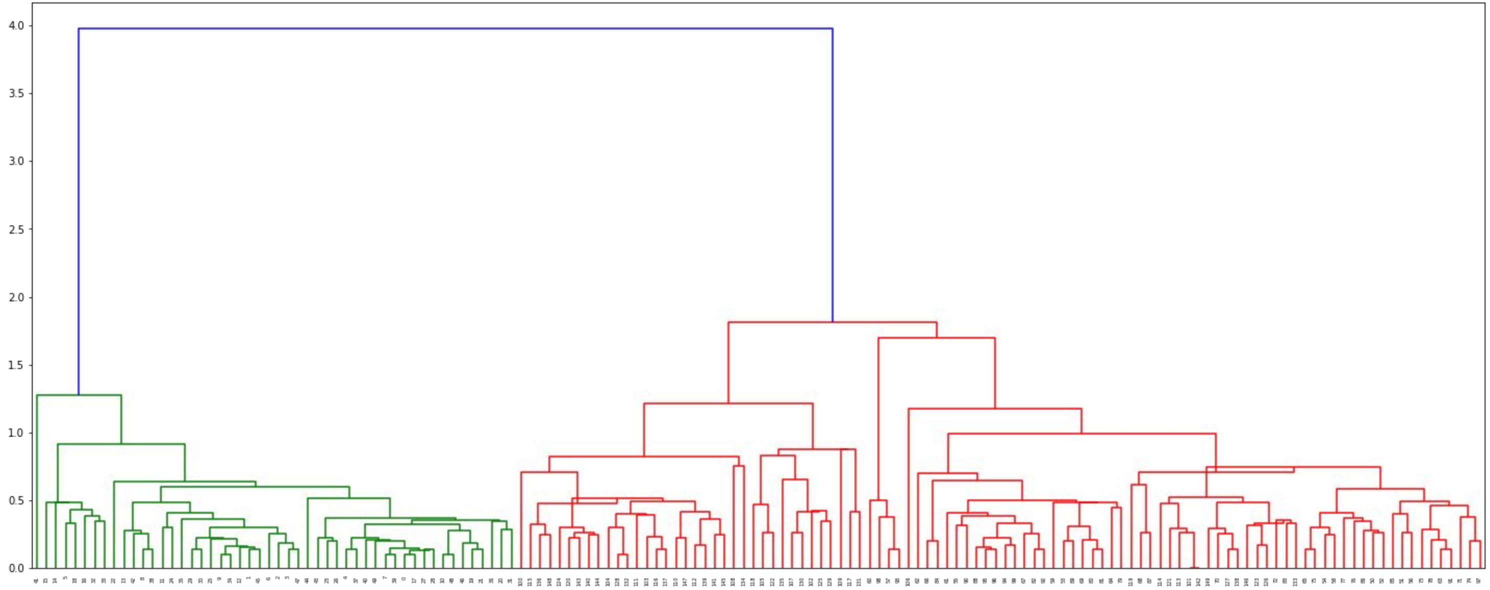
source

Hierarchical Clustering

- We don’t know the number of clusters up front
- With hierarchical clustering, we can choose the end number of clusters after we create the tree.
- Two different methods
  - Agglomerative clustering (bottom-up approach):we start with each data point as a cluster, and we group clusters based on similarity.
  - Divisive clustering (top-down approach): split recursively until
- Does not scale well for larger or high-dimensional data sets (computationally expensive)

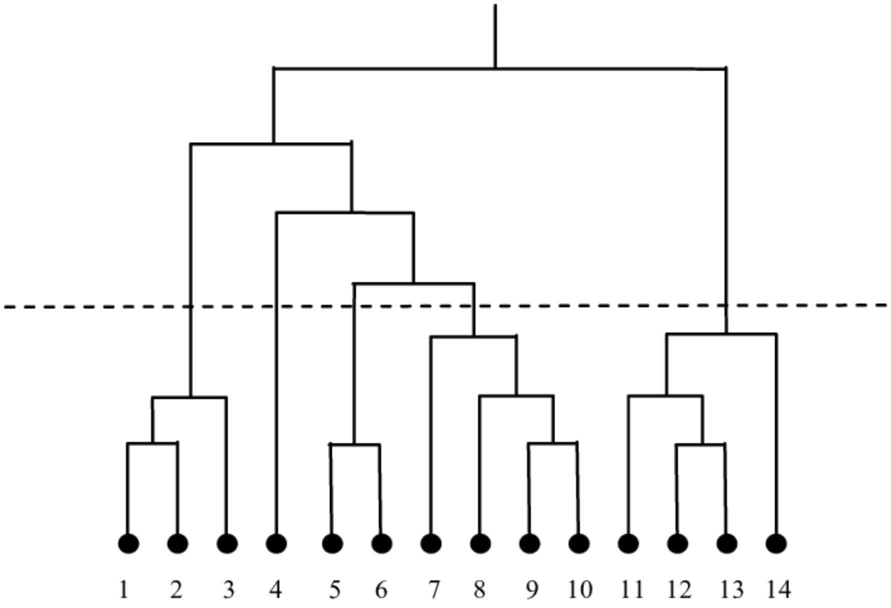
Dendrogram

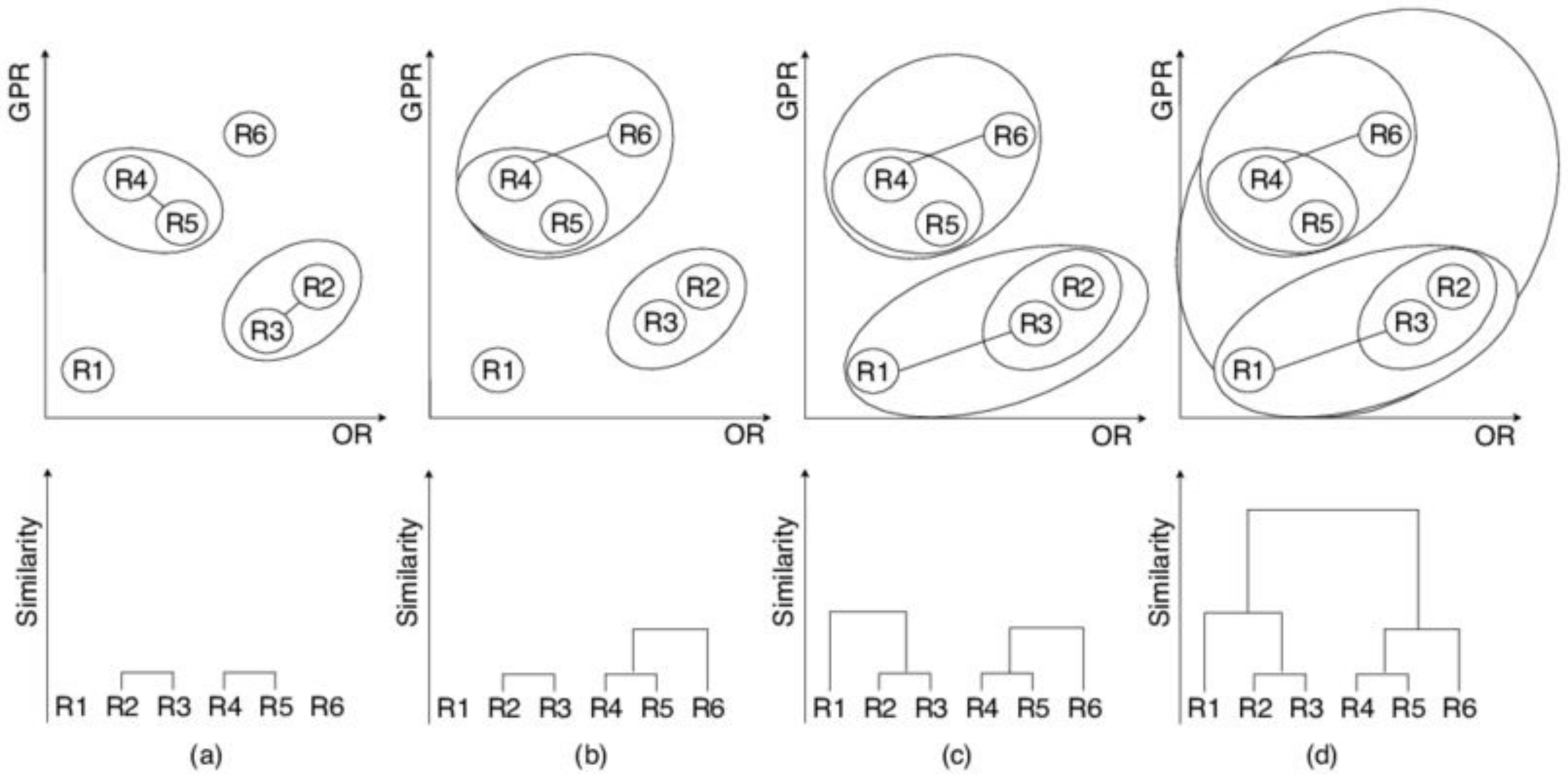
- Using a tree structure called a dendrogram.
- The height on the dendrogram where two groups are linked gives us the linkage order
- Height of the links in indicate the amount of dissimilarity between the clusters



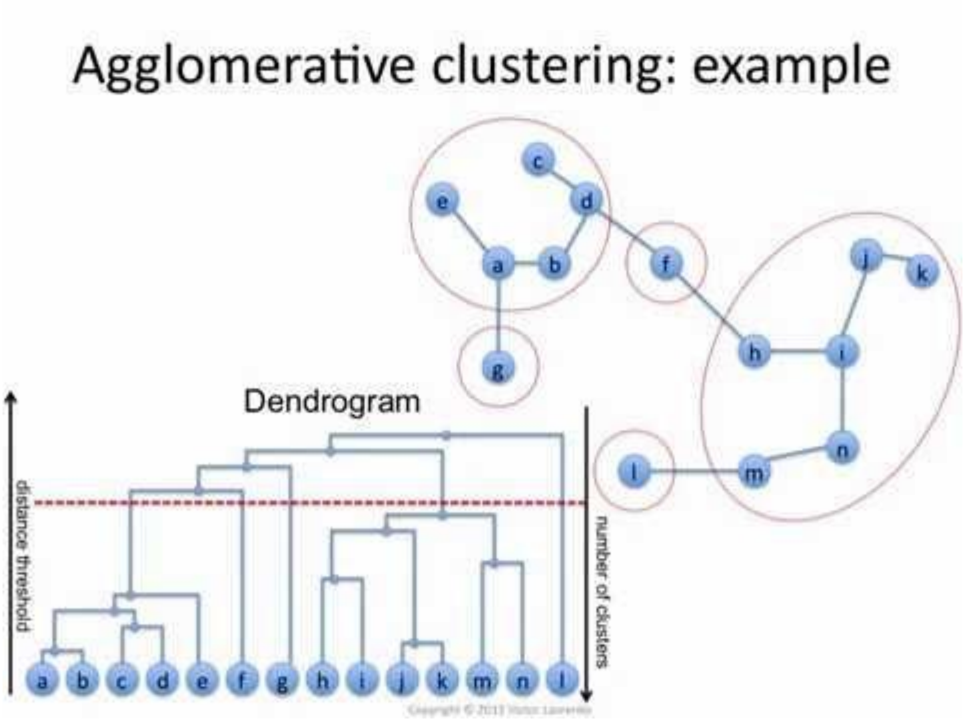
Dendrogram

- We can determine the number of clusters by making a cut in the dendrogram





https://www.researchgate.net/figure/Hierarchical-clustering-steps_fig9_349620146

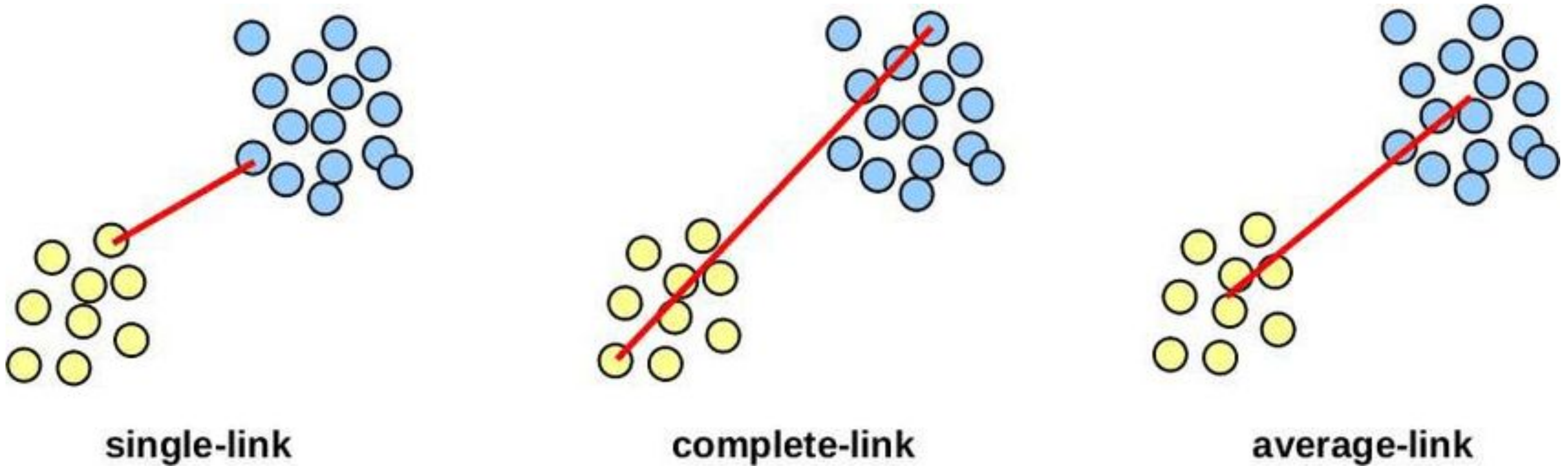


https://youtu.be/XJ3194AmH40

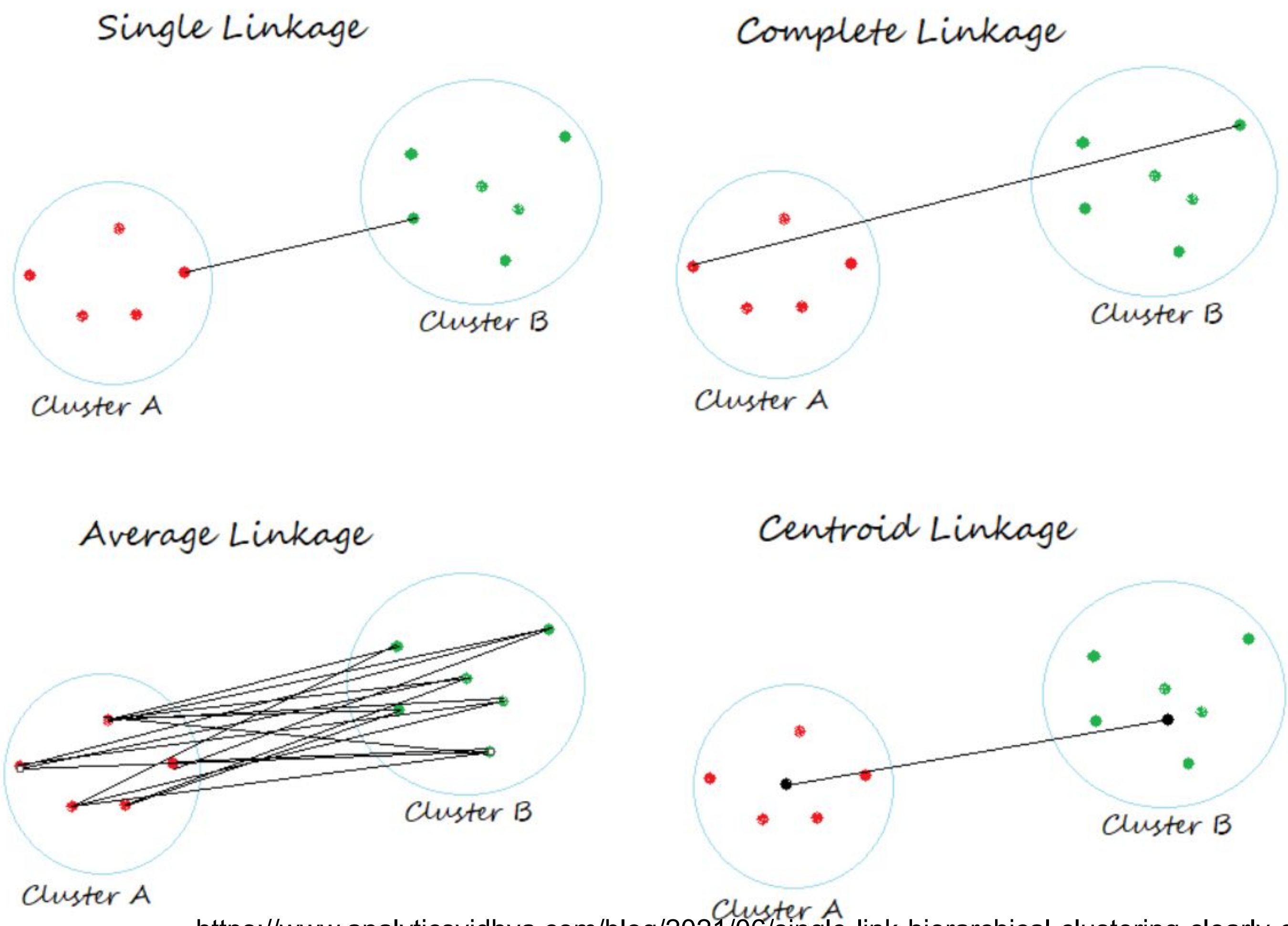
Linkage

- How to decide which clusters to join together in hierarchical clustering?
- We need to define a dissimilarity measure
  - **Complete (Maximal) Linkage:** The furthest distance between an element of A and an element of B
  - **Single (Minimal) Linkage:** The shortest distance between an element of A and an element of B
  - **Average Linkage:** The average distance between any element of A and any element of B
  - **Centroid Linkage:** The distance between the centroid of A and the centroid of B
- The first 3 methods are computationally expensive, O(|A||B|) per linkage computation.
- Centroid Linkage method suffers from inversion

Linkage



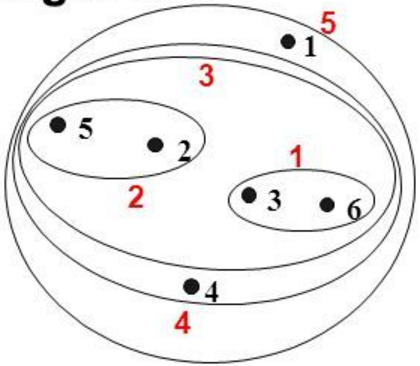
https://www.researchgate.net/figure/The-three-linkage-types-of-hierarchical-clustering-single-link-complete-link-and_fig57_281014334



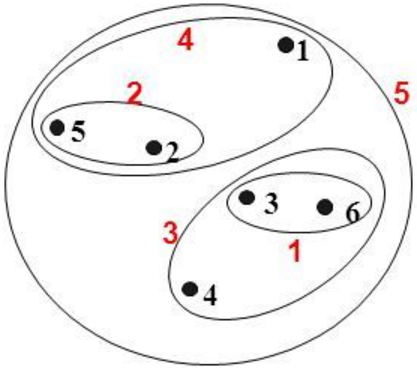
https://www.analyticsvidhya.com/blog/2021/06/single-link-hierarchical-clustering-clearly-explained/

Hierarchical Clustering: Comparison

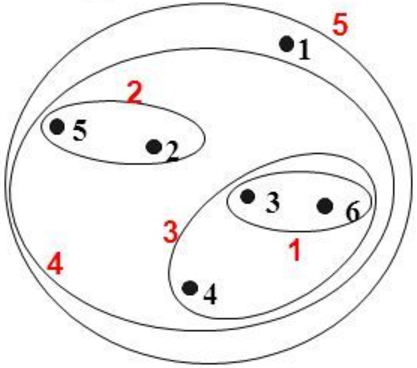
Single-link



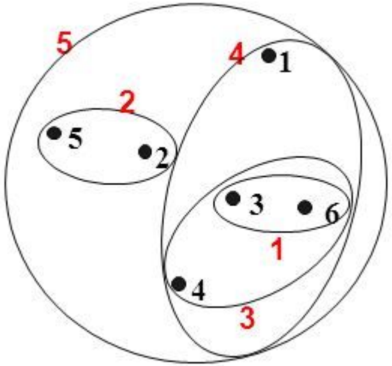
Complete-link



Average-link

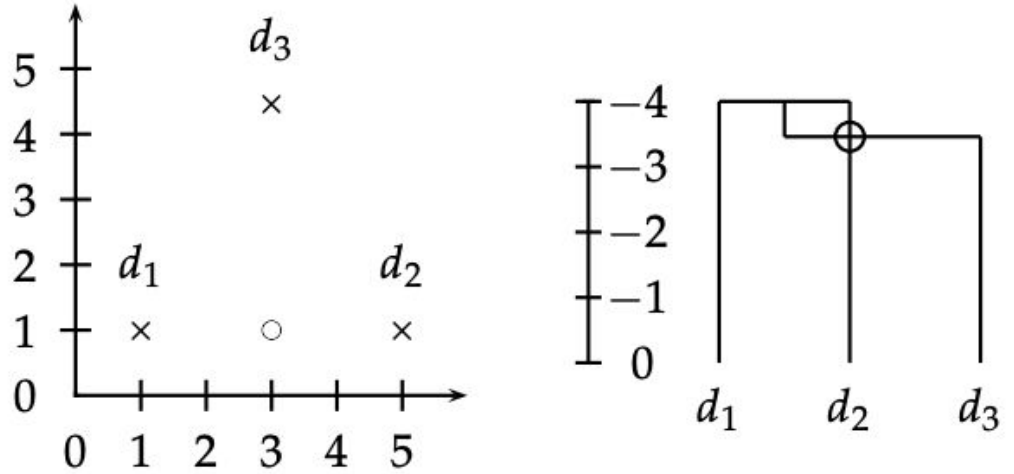


Centroid distance



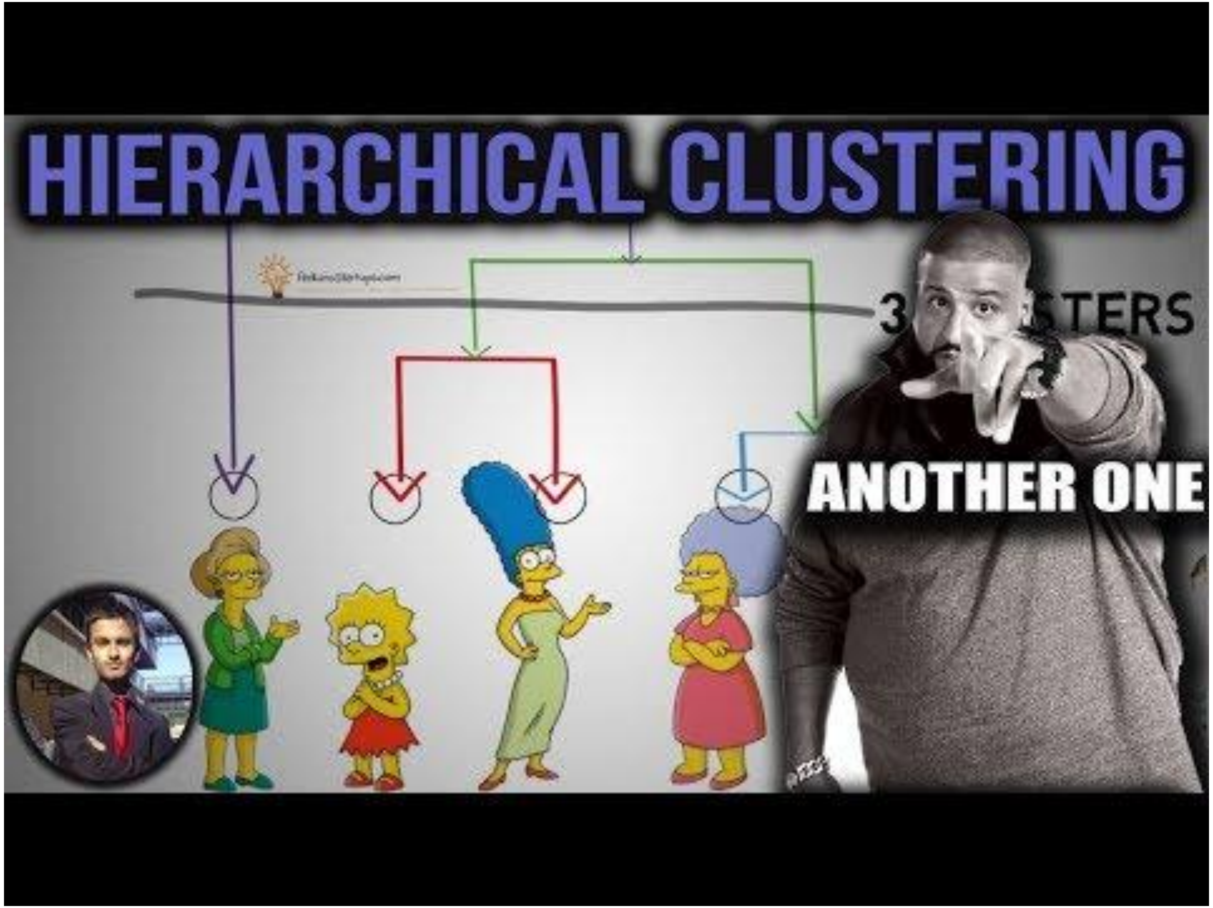
Distance measure

- In addition to the type of linkage we need a distance measure as well
  - Euclidean distance
  - Minkowski distances
  - Chebyshev distance
  - Manhattan distance
  - Hamming distance (mismatches between words)
  - Tempo/rhythm similarity (for songs)



► **Figure 17.12** Centroid clustering is not monotonic. The documents d_1 at $(1+\epsilon, 1)$, d_2 at $(5, 1)$, and d_3 at $(3, 1+2\sqrt{3})$ are almost equidistant, with d_1 and d_2 closer to each other than to d_3. The non-monotonic inversion in the hierarchical clustering of the three points appears as an intersecting merge line in the dendrogram. The intersection is circled.

https://nlp.stanford.edu/IR-book/pdf/17hier.pdf https://youtu.be/EUQY3hL38cw




Agglomerative Hierarchical Clustering
Solved Example
18
22
25
27
42
43
Subscribe to Mahesh Huddar
Visit: vtupulse.com